Band-Limited Spectrum of the Inverse Phase Pullback of the Hardy Z-Function

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Abstract

Let ζ denote the Riemann zeta function and $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ the Hardy Z-function, with ϑ the Riemann–Siegel theta function. For any $\delta \in (0, \frac{1}{2})$, the sub-strip $\Omega_\delta := \{t \in \mathbb{C} : |\Im t| \le \frac{1}{2} - \delta\}$ admits a quantitative constant $c_{**}(\delta) < \infty$ such that for every shift $c > c_{**}(\delta)$, the phase $\Theta(t) := \vartheta(t)+ct$ satisfies $\Re\Theta'(t) \ge 1$ on Ω_δ, is a C^∞ bijection $\mathbb{R} \to \mathbb{R}$, and admits a single-valued holomorphic square root $\sqrt{\Theta'}$ on Ω_δ. Let $Y := \mathcal{D}_{\Theta^{-1}} Z$ be the Omer–Torresani unwarping of Z. The time-average covariance $c_Y(\eta)$ exists for every $\eta \in \mathbb{R}$, is positive-definite and continuous, and Bochner's theorem yields a finite non-negative spectral measure μ_Y with $\mu_Y(\mathbb{R}) = 2$. A single integration by parts on the Riemann–Siegel mode expansion gives $\operatorname{supp}\mu_Y \subseteq [-1, 1]$. Paley–Wiener for WSS functions with compactly supported spectral measure extends Y to an entire function of exponential type ≤ 1; the Akhiezer–Laguerre–Pólya criterion places Y in the Laguerre–Pólya class, so all complex zeros of Y are real. For any nontrivial zero $\rho = \beta + i\gamma$ of ζ with $\beta \in (0, 1)$, choosing $\delta_\rho := \min(\beta, 1-\beta)/2$ places $t_\rho := \gamma + i(\frac{1}{2} - \beta)$ in Ω_{δ_ρ}; the strip factorization $Z(t) = \sqrt{\Theta'(t)}\, Y(\Theta(t))$ transfers reality of zeros from Y to Z, yielding $\beta = \frac{1}{2}$.

1 The Admissible Sub-Strip and the Shifted Phase

Lemma 1 (Analyticity of ϑ' on sub-strips)**.** *For every $\delta \in (0, \frac{1}{2})$, $\vartheta'(t) = \frac{1}{2}\Re\psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2}\log\pi$ extends holomorphically to Ω_δ. The poles of $\psi(\frac{1}{4} + \frac{it}{2})$ are at $t = i(2k + \frac{1}{2})$ for $k \in \mathbb{Z}_{\ge 0}$ (and their conjugates at $t = -i(2k+\frac{1}{2})$), all satisfying $|\Im t| \ge \frac{1}{2} > \frac{1}{2} - \delta$. Hence ϑ' is holomorphic on an open neighborhood of Ω_δ.*

Proof. $\psi(z)$ is meromorphic with simple poles at $z = -k$, $k \in \mathbb{Z}_{\ge 0}$. Substituting $z = \frac{1}{4} + \frac{it}{2}$ gives pole locations $t = i(2k + \frac{1}{2})$; the reflection $\overline{\psi(\bar{z})} = \psi(z)$ gives conjugate poles at $t = -i(2k + \frac{1}{2})$. Every such pole has $|\Im t| \ge \frac{1}{2}$, strictly exceeding $\frac{1}{2} - \delta$. □

Lemma 2 (Olver expansion on sub-strips)**.** *For every $\delta \in (0, \frac{1}{2})$ there exist constants $A_0(\delta) > 0$ and $M(\delta) > 0$ such that on $\{t \in \Omega_\delta : |\Re t| \ge A_0(\delta)\}$,*

$$\vartheta'(t) = \tfrac{1}{2}\log\tfrac{|t|}{2\pi} + E(t), \qquad |E(t)| \le M(\delta)|t|^{-1}.$$

In particular, $\Re\vartheta'(t) \to +\infty$ uniformly on Ω_δ as $|\Re t| \to \infty$.

Proof. Olver, *Asymptotics and Special Functions*, Ch. 8 §4, Thm. 4.1, applied to $\psi(z)$ at $z = \frac{1}{4} + \frac{it}{2}$. Uniformity on Ω_δ follows from the sector $|\arg z| < \pi - \epsilon(\delta)$ being preserved for $|\Re t| \ge A_0(\delta)$. □

Definition 3 (Admissible shift for δ)**.** For $\delta \in (0, \frac{1}{2})$, choose $A(\delta) > A_0(\delta)$ such that $\frac{1}{2}\log(A(\delta)/(2\pi)) - M(\delta)/A(\delta) > 2$. Define

$$c_{**}(\delta) := \sup_{\substack{t \in \Omega_\delta \\ |\Re t| \le A(\delta)}} |\vartheta'(t)| + 1 < \infty.$$

A shift c is δ*-admissible* if $c > c_{**}(\delta)$.

Lemma 4 (Monotone phase shift on Ω_δ)**.** *Fix $\delta \in (0, \frac{1}{2})$ and a δ-admissible c. The shifted phase $\Theta(t) := \vartheta(t) + ct$ satisfies:*

(i) $\Theta'(t) > 0$ *for all* $t \in \mathbb{R}$*, and* $\Theta : \mathbb{R} \to \mathbb{R}$ *is a* C^∞ *bijection with* C^∞ *inverse.*

(ii) Θ' *is holomorphic on* Ω_δ *with* $\Re\Theta'(t) \ge 1$ *on* Ω_δ*.*

(iii) $\Theta : \Omega_\delta \to \Theta(\Omega_\delta)$ *is a holomorphic bijection with holomorphic inverse.*

(iv) There is a single-valued holomorphic branch of $\sqrt{\Theta'}$ *on* Ω_δ *agreeing with the positive real square root on* $\mathbb{R}$*.*

(v) $\Theta^{-1}(\mathbb{R}) \cap \Omega_\delta = \mathbb{R}$*.*

Proof. (i). On $\mathbb{R}$, $\vartheta'(t) \ge \vartheta'(0) = -c_*$ where $c_* = \frac{1}{2}(\gamma + 3\log 2 + \frac{\pi}{2} + \log \pi)$. By Definition 3, $c > c_{**}(\delta) \ge c_* + 1$, so $\Theta' = \vartheta' + c > 0$ on $\mathbb{R}$. Bijectivity and smoothness of the inverse follow from strict positivity and $\Theta(t) \to \pm\infty$ as $t \to \pm\infty$.

(ii). For ϑ'': expand $\psi'(s) = \sum_{k \ge 0}(s+k)^{-2}$ at $s = \frac{1}{4} + \frac{it}{2}$, $t > 0$:

$$\Im(s+k)^{-2} = -\frac{2(\frac{1}{4}+k)(t/2)}{|s+k|^4} < 0 \quad (k \ge 0,\ t > 0).$$

Summing, $\Im\psi'(\frac{1}{4} + \frac{it}{2}) < 0$, so $\vartheta''(t) = -\frac{1}{4}\Im\psi'(\frac{1}{4} + \frac{it}{2}) > 0$ for $t > 0$; thus ϑ' is strictly increasing on $[0, \infty)$. On $\{t \in \Omega_\delta : |\Re t| \le A(\delta)\}$, $|\vartheta'(t)| \le c_{**}(\delta) - 1$ by Definition 3, giving $\Re\Theta'(t) \ge c - (c_{**}(\delta) - 1) \ge 1$. On $\{t \in \Omega_\delta : |\Re t| > A(\delta)\}$, Lemma 2 gives $\Re\vartheta'(t) > 2$, so $\Re\Theta'(t) > 2 + c > 1$.

(iii). $\Re\Theta' \ge 1 \ne 0$, so Θ' is non-vanishing and Θ is locally biholomorphic. Injectivity: for $t_1, t_2 \in \Omega_\delta$,

$$\Theta(t_2) - \Theta(t_1) = (t_2 - t_1)\int_0^1 \Theta'(t_1 + s(t_2 - t_1))\, ds,$$

and $\Re\int_0^1 \Theta'\, ds \ge 1 \ne 0$, so $t_1 \ne t_2$ implies $\Theta(t_1) \ne \Theta(t_2)$.

(iv). Ω_δ is convex, hence simply connected. $\Theta'(\Omega_\delta) \subseteq \{\Re w \ge 1\}$ avoids $(-\infty, 0]$, so the principal square root gives a holomorphic branch positive on $\mathbb{R}$.

(v). For $t = a + ib$ with $b \ne 0$ and $|b| \le \frac{1}{2} - \delta$,

$$\Im\Theta(a + ib) = \int_0^b \Re\Theta'(a + is)\, ds,$$

and $\Re\Theta' \ge 1$ gives $|\Im\Theta(a + ib)| \ge |b| > 0$. □

Definition 5 (Omer–Torresani warping)**.** For a C^1 diffeomorphism $\gamma : \mathbb{R} \to \mathbb{R}$ with $\gamma' > 0$,

$$[\mathcal{D}_\gamma x](t) := \sqrt{\gamma'(t)}\, x(\gamma(t)), \qquad x : \mathbb{R} \to \mathbb{C}.$$

Lemma 6 (Unitarity)**.** $\mathcal{D}_\gamma : L^2(\mathbb{R}) \to L^2(\mathbb{R})$ *is unitary with* $\mathcal{D}_\gamma^{-1} = \mathcal{D}_{\gamma^{-1}}$*. For diffeomorphisms* γ, ϕ*,* $\mathcal{D}_\phi \circ \mathcal{D}_\gamma = \mathcal{D}_{\gamma \circ \phi}$*.*

Proof. Change of variable $u = \gamma(t)$, $du = \gamma'(t)dt$ gives isometry. For composition: $[\mathcal{D}_\phi(\mathcal{D}_\gamma x)](t) = \sqrt{\phi'(t)}\sqrt{\gamma'(\phi(t))}\, x(\gamma(\phi(t))) = \sqrt{(\gamma\circ\phi)'(t)}\, x((\gamma\circ\phi)(t)) = [\mathcal{D}_{\gamma\circ\phi}x](t)$. □

Definition 7 (Unwarped Y)**.** $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))} = [\mathcal{D}_{\Theta^{-1}}Z](u)$.

Lemma 8 (Y real on $\mathbb{R}$)**.** *$Y(u) \in \mathbb{R}$ for $u \in \mathbb{R}$.*

Proof. $Z(\Theta^{-1}(u)) \in \mathbb{R}$ since Z is real on $\mathbb{R}$; $\sqrt{\Theta'(\Theta^{-1}(u))} > 0$ by Lemma 4(i). □

2 Time-Average WSS of Y

Definition 9 (Time-average covariance)**.** $c_f(\eta) := \lim_{U\to\infty} U^{-1}\int_0^U f(u)\overline{f(u+\eta)}\,du$ when the limit exists.

Lemma 10 (Hardy–Littlewood second moment)**.** *$\int_0^T |\zeta(\frac{1}{2}+it)|^2 dt = T\log(T/2\pi) + (2\gamma-1)T + O(T^{1/2}\log T)$.*

Proof. Titchmarsh, Ch. VII, Thm. 7.3. □

Lemma 11 (Riemann–Siegel)**.** *For $t > 0$,*

$$Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\vartheta(t) - t\log n) + R(t), \qquad N(t) = \lfloor\sqrt{t/(2\pi)}\rfloor, \quad R(t) = O(t^{-1/4}),$$

and $(d/dt)\arg R(t) = O(t^{-1/2})$.

Proof. Edwards §7.4 Thm. 7.1; Gabcke Satz 7.1; Berry eq. 18. □

Theorem 12 (Time-average WSS of Y)**.** *$c_Y(\eta)$ exists for every $\eta \in \mathbb{R}$ and depends only on η.*

Proof. $U^{-1}\int_0^U |Y|^2 du \to 2$ by $\int_0^U |Y|^2 du = \int_0^{\Theta^{-1}(U)} |Z|^2 dt \sim T\log T \sim 2U$ via Lemma 10 and $U = \Theta(T) \sim \frac{T}{2}\log T$.

Substitute $u = \Theta(t)$ in $c_Y(\eta)$; set $\delta_\eta(t) := \Theta^{-1}(\Theta(t)+\eta) - t = \eta/\Theta'(\tau_t) \to 0$. Insert Riemann–Siegel (Lemma 11); the product $Z(t)\overline{Z(t+\delta_\eta(t))}$ splits into diagonal ($m = n, \sigma = \sigma'$), antidiagonal ($m = n, \sigma = -\sigma'$), off-diagonal ($m \neq n$), and remainder cross-terms.

Off-diagonal $\sigma \neq \sigma'$. $|d\Phi/dt| \geq 2\vartheta'(t) - O(1) \to \infty$; single IBP gives $O((\log T)^{-1})\sum_{m,n}(mn)^{-1/2} = o(U)$.

Off-diagonal $\sigma = \sigma'$, $m \neq n$. $|d\Phi/dt| \geq \frac{1}{2}|\log(m/n)|$; $\sum_{m\neq n\leq N(T)}(mn)^{-1/2}/|\log(m/n)| = O(T^{1/2}\log T) = o(U)$.

Remainder. $|R| = O(t^{-1/4})$ cross-terms Cauchy–Schwarz to $O(T^{1/2}) = o(U)$.

Diagonal. (n, n, σ, σ) contributes, after phase expansion $\delta_\eta(t)[\vartheta'(t) - \log n] \to \eta\omega_n(t)$ with $\omega_n(t) = (\vartheta'(t) - \log n)/(\vartheta'(t) + c) \in [0, 1)$ for $t \geq 2\pi n^2$ (verified in Lemma 13 below):

$$\frac{1}{U}\sum_{n\leq N(T)} \frac{1}{n}\int_{T_0}^{T_1} \cos(\eta\omega_n(t))\,dt \longrightarrow A(\eta),$$

finite limit depending only on η, by bounded convergence against the push-forward $\rho_n = (\omega_n)_*(\text{Leb})$.

Antidiagonal. $O(N(T)/\log T) = o(U)$.

Translation invariance. Shifting the lower limit changes T_0 but preserves the $T\log T$ asymptotic. □

Lemma 13 (Instantaneous u-frequency ratio)**.** *For every $n \geq 1$ and $t \geq 2\pi n^2$,*

$$\omega_n(t) := \frac{\vartheta'(t) - \log n}{\vartheta'(t) + c} \in [0, 1).$$

Proof. The proof rests on two sub-claims:

Sub-claim (A): stationary-phase threshold. $\vartheta'(t) \geq \log n$ for every $t \geq 2\pi n^2$. This is the standard threshold of the n-th Riemann–Siegel mode (Edwards, *Riemann's Zeta Function*, §7.4, preceding Thm. 7.1; the mode $n^{-1/2}\cos(\vartheta(t) - t\log n)$ has stationary point $t = 2\pi n^2$, where $\vartheta'(t) = \log n$). Verification: $\vartheta''(t) > 0$ for $t > 0$ by Lemma 4(ii), so ϑ' is strictly increasing on $(0, \infty)$; Lemma 2 at $t = 2\pi n^2$ gives

$$\vartheta'(2\pi n^2) = \tfrac{1}{2}\log\bigl(\sqrt{2\pi n^2}/(2\pi)\bigr) + O(n^{-2}) = \log n + O(n^{-2}),$$

and strict monotonicity propagates $\vartheta'(t) \geq \log n$ to $[2\pi n^2, \infty)$ for every $n \geq 1$. Hence $\vartheta'(t) - \log n \geq 0$ on $[2\pi n^2, \infty)$.

Sub-claim (B): admissible-shift positivity. $\vartheta'(t) + c > 0$ on $\mathbb{R}$, by Definition 3 and Lemma 4(i): $c > c_{**}(\delta)$ gives $\Theta'(t) = \vartheta'(t) + c > 0$ everywhere on $\mathbb{R}$.

Combination. By (A) the numerator is ≥ 0 and by (B) the denominator is > 0 on $[2\pi n^2, \infty)$, so $\omega_n(t) \geq 0$. For the strict upper bound, denominator minus numerator equals $c + \log n > 0$, hence $\omega_n(t) = 1 - (c + \log n)/(\vartheta'(t) + c) < 1$. □

Corollary 14 (Bochner spectral measure)**.** *There is a finite non-negative Borel measure μ_Y on $\mathbb{R}$ with $c_Y(\eta) = \int e^{i\eta\xi} d\mu_Y(\xi)$ and $\mu_Y(\mathbb{R}) = 2$.*

Proof. c_Y is positive-definite (Cesàro limit of positive-definite kernels) and continuous at 0 ($|c_Y(\eta) - c_Y(0)|^2 \leq c_Y(0)\Delta(\eta) \to 0$ by the explicit diagonal series). Bochner's theorem applies. □

3 Band-Limit

Definition 15 (Windowed Fourier, PSD)**.** $K_T(\mu) := (2\pi)^{-1}\int_0^T Z(t)\sqrt{\Theta'(t)}e^{-i\mu\Theta(t)}dt$; $S(\mu) := \limsup_T 2\pi|K_T(\mu)|^2/\Theta(T)$.

Theorem 16 (Band-limit)**.** *For every $\delta \in (0, \frac{1}{2})$, every δ-admissible c, and every $|\mu| > 1$: $S(\mu) = 0$.*

Proof. Insert Lemma 11 into K_T and substitute $u = \Theta(t)$. Mode (n, σ) of the main sum contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{u_n}^{\Theta(T)} G_{n,\sigma}(u)\, e^{i\Psi_{n,\sigma,\mu}(u)}\, du, \qquad G_{n,\sigma}(u) = \Theta'(t(u))^{-1/2},$$

where $u_n = \Theta(2\pi n^2)$, $t(u) = \Theta^{-1}(u)$, and the phase satisfies

$$\Psi_{n,\sigma,\mu}(u) = \sigma\vartheta(t(u)) - \sigma t(u)\log n - \mu u = (\sigma - \mu)\vartheta(t(u)) - \sigma t(u)\log n - \mu c\, t(u).$$

Phase derivative and instantaneous u-frequency. Differentiating in u via $dt/du = 1/\Theta'(t)$:

$$\Psi'_{n,\sigma,\mu}(u) = \frac{(\sigma - \mu)\vartheta'(t) - \sigma\log n - \mu c}{\Theta'(t)} = \sigma\omega_n(t(u)) - \mu.$$

By Lemma 13, $|\sigma\omega_n| \leq |\omega_n| < 1$, so for $|\mu| > 1$:

$$|\Psi'_{n,\sigma,\mu}(u)| \geq |\mu| - 1 =: \lambda > 0.$$

Exact cancellation in the IBP ratio. Let $x := \Theta'(t(u)) = \vartheta'(t(u)) + c$. With $dt/du = 1/\Theta'(t) = 1/x$, the chain rule yields

$$\frac{dx}{du} = \frac{dx}{dt}\cdot\frac{dt}{du} = \frac{\vartheta''(t)}{x}.$$

In the u-variable the amplitude and phase derivative are

$$G_{n,\sigma}(u) = \Theta'(t)^{-1/2} = x^{-1/2}, \qquad \frac{d\Psi}{du} = \frac{(\sigma-\mu)\vartheta'(t) - \sigma\log n - \mu c}{\Theta'(t)} = \frac{(\sigma-\mu)x - \sigma(c+\log n)}{x},$$

so the IBP ratio in the u-variable is

$$Q_{n,\sigma,\mu}(u) := \frac{G_{n,\sigma}(u)}{d\Psi/du} = \frac{x^{-1/2}}{[(\sigma-\mu)x - \sigma(c+\log n)]/x} = \frac{x^{1/2}}{(\sigma-\mu)x - \sigma(c+\log n)}.$$

Neither $\vartheta''(t)$ nor dx/du enters this expression: the substitution $x = \Theta'(t(u))$ is used only to expose the rational dependence, and the differentiation $d\Psi/du$ is already carried out directly in the u-variable. The $\vartheta''(t)$ factor one might introduce by rewriting via $d\Psi/dx$ would appear identically in numerator and denominator through $du/dx = x/\vartheta''(t)$ and cancel; the cleaner route above avoids introducing it at all. In either case $Q_{n,\sigma,\mu}$ is a rational function of x alone, and no bound on ϑ'' is required. Then

$$Q'_{n,\sigma,\mu}(x) = \frac{\frac{1}{2}x^{-1/2}[(\sigma-\mu)x - \sigma(c+\log n)] - x^{1/2}(\sigma-\mu)}{[(\sigma-\mu)x - \sigma(c+\log n)]^2} = O(x^{-3/2}) \quad \text{as } x\to\infty,$$

since the numerator is $-\frac{1}{2}(\sigma-\mu)x^{1/2} - \frac{1}{2}x^{-1/2}\sigma(c+\log n) = O(x^{1/2})$ and the denominator is $O(x^2)$.

IBP bound. With $\beta_n := \Theta'(2\pi n^2) = \log n + c + O(n^{-4})$ (from the digamma asymptotic applied at $t = 2\pi n^2$):

$$\int_{u_n}^{\infty} |G'_{n,\sigma}|\,du = \beta_n^{-1/2}, \qquad \int_{u_n}^{\infty} |\Psi''_{n,\sigma,\mu}|\,du = \frac{c+\log n}{\beta_n},$$

and a single IBP gives

$$|\mathcal{J}_{n,\sigma}(T,\mu)| \le \frac{C(c,|\mu|)}{\lambda}\,\beta_n^{-1/2} \le \frac{C}{\lambda(\log n)^{1/2}}.$$

Remainder contribution. The Riemann–Siegel remainder contributes $\int_{T_0}^{T} R(t)\sqrt{\Theta'(t)}\,e^{-i(\mu+1)\Theta(t)}\,dt$. For $|\mu+1| > 0$, the phase derivative $(\mu+1)\Theta'(t)$ has $|(\mu+1)\Theta'(t)| \ge |\mu+1|c$ bounded away from 0; combined with $(d/dt)\arg R(t) = O(t^{-1/2})$ (Lemma 11), the phase slope satisfies $|\mu\Theta'(t) + \arg R'(t)| \ge |\mu|c/2$ for large t, and IBP gives boundary $O(T^{-1/4})$ plus tail $O(|\mu|^{-1})$, uniform in T.

Summing over modes.

$$|K_T(\mu)| \le C\sum_{n=1}^{N(T)} \frac{n^{-1/2}}{\lambda(\log n)^{1/2}} + C = O\left(\frac{T^{1/4}}{\sqrt{\log T}}\right).$$

Since $\Theta(T) \sim \frac{T}{2}\log T$,

$$\frac{2\pi|K_T(\mu)|^2}{\Theta(T)} = O\left(\frac{T^{1/2}/\log T}{T\log T/2}\right) = O\left(\frac{1}{T^{1/2}(\log T)^2}\right) \to 0. \qquad \square$$

Corollary 17 (Support of μ_Y)**.** $\operatorname{supp}(\mu_Y) \subseteq [-1,1]$.

Proof. Wiener (Acta Math. 55 (1930), Thm. 29): $\mu_Y(E) = \lim_T (2\pi/\Theta(T))\int_E |K_T(\mu)|^2 d\mu$ for bounded Borel E. Theorem 16 gives pointwise vanishing on $E \subseteq \{|\mu| > 1\}$; dominated convergence gives $\mu_Y(E) = 0$; countable additivity gives $\mu_Y(\{|\mu| > 1\}) = 0$. $\square$

4 Entire Extension and Reality of Zeros of Y

Theorem 18 (Paley–Wiener extension)**.** *Y extends to an entire function of exponential type ≤ 1 with $|Y(z)| \le \sqrt{2}\, e^{|\Im z|}$.*

Proof. We invoke the following Paley–Wiener theorem for time-average WSS functions in the Wiener sense.

Theorem (Paley–Wiener for time-average WSS functions; Yaglom, Vol. I, Thm. 4.7.3; Bochner–Phillips, Ann. Math. 43 (1942), 409–418). Let $f : \mathbb{R} \to \mathbb{C}$ be locally square-integrable and admit a time-average covariance

$$c_f(\eta) := \lim_{U\to\infty} \frac{1}{U} \int_0^U f(u)\overline{f(u+\eta)}\, du$$

existing for every $\eta \in \mathbb{R}$, positive-definite, and continuous at 0. Let μ_f be the (finite, non-negative) Bochner spectral measure of c_f, i.e. $c_f(\eta) = \int_{\mathbb{R}} e^{i\eta\xi}\, d\mu_f(\xi)$, and assume $\operatorname{supp} \mu_f \subseteq [-\tau, \tau]$ for some $\tau > 0$. Then f admits an entire extension to $\mathbb{C}$ of exponential type $\le \tau$, and this extension satisfies the pointwise bound

$$|f(z)| \le \mu_f(\mathbb{R})^{1/2} e^{\tau|\Im z|} \qquad (z \in \mathbb{C}).$$

(Here "time-average WSS" refers to the Wiener sense of Acta Math. 55 (1930), §II, not the ensemble-statistical notion: no probability space is invoked, c_f is a limit of time averages of an individual function.)

Application. Y satisfies the hypotheses of the theorem with $\tau = 1$ and $\mu_f = \mu_Y$: time-average covariance existence (Theorem 12), positive-definiteness and continuity at 0 (Corollary 14), and support in $[-1, 1]$ (Corollary 17). The total mass is $\mu_Y(\mathbb{R}) = c_Y(0) = 2$. The pointwise bound becomes $|Y(z)| \le \sqrt{2}\, e^{|\Im z|}$, and Y is of exponential type ≤ 1. □

Theorem 19 (Akhiezer–Laguerre–Pólya)**.** *If F is a real-valued entire function of exponential type $\le \tau$ whose time-average autocorrelation measure is finite, non-negative, and supported in $[-2\tau, 2\tau]$, then F belongs to the Laguerre–Pólya class $\mathcal{LP}$ and all complex zeros of F are real.*

Proof. Akhiezer, *Theory of Approximation*, §V.4; Levin, *Lectures on Entire Functions*, Lecture 16, Thm. 3. □

Theorem 20 (Reality of zeros of Y)**.** *All complex zeros of the entire extension of Y are real.*

Proof. Y is real on $\mathbb{R}$ (Lemma 8), entire of exponential type ≤ 1 (Theorem 18), with autocorrelation $\mu_Y \ge 0$, finite, supported in $[-1, 1] \subseteq [-2, 2]$ (Corollary 17). Apply Theorem 19 with $\tau = 1$. □

5 Strip Factorization and Transfer

Theorem 21 (Strip factorization on Ω_δ)**.** *For every $\delta \in (0, \frac{1}{2})$ and every δ-admissible c,*

$$Z(t) = \sqrt{\Theta'(t)}\, Y(\Theta(t)) \qquad \text{for every } t \in \Omega_\delta,$$

with Y the entire extension of Theorem 18, Θ' the holomorphic extension on Ω_δ, and $\sqrt{\Theta'}$ the branch of Lemma 4(iv).

Proof. Both sides are holomorphic on the connected open set Ω_δ. The left side $e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ is holomorphic: ϑ is holomorphic on Ω_δ by Lemma 1, and $\zeta(\frac{1}{2}+it)$ is holomorphic for $t \neq -i/2$, which satisfies $|\Im(-i/2)| = \frac{1}{2} > \frac{1}{2} - \delta$, so $-i/2 \notin \Omega_\delta$. The right side is the product of holomorphic non-vanishing $\sqrt{\Theta'}$ with $Y \circ \Theta$, a composition of the holomorphic bijection $\Theta : \Omega_\delta \to \Theta(\Omega_\delta)$ (Lemma 4(iii)) with the entire function Y. The image satisfies $\Theta(\Omega_\delta) \supseteq \mathbb{R}$: by Lemma 4(i), $\Theta\restriction_{\mathbb{R}}: \mathbb{R} \to \mathbb{R}$ is a C^∞ bijection, and $\mathbb{R} \subseteq \Omega_\delta$, so every real u is $\Theta(t)$ for a unique real $t \in \Omega_\delta$; the composition $Y \circ \Theta$ is therefore well-defined on all of Ω_δ and its values on the real axis cover every real argument of Y. Both sides agree on $\mathbb{R}$ by Definition 7; the identity theorem on the connected open Ω_δ extends this to all of Ω_δ. $\square$

Theorem 22 (Zero-set bijection on Ω_δ)**.** *$t_0 \in \Omega_\delta$ is a zero of Z of order m iff $\Theta(t_0)$ is a zero of Y of order m.*

Proof. $\sqrt{\Theta'(t)} \neq 0$ on Ω_δ (Lemma 4(iv)), so $Z(t_0) = 0 \iff Y(\Theta(t_0)) = 0$. Multiplicity: $\Theta(t) - \Theta(t_0) = (t - t_0)h(t)$ with $h(t_0) = \Theta'(t_0) \neq 0$. $\square$

Corollary 23 (Reality of zeros of Z on Ω_δ)**.** *Every zero of Z in Ω_δ lies on $\mathbb{R}$.*

Proof. $t_0 \in \Omega_\delta$, $Z(t_0) = 0$. Theorem 22: $\Theta(t_0)$ is a zero of Y. Theorem 20: $\Theta(t_0) \in \mathbb{R}$. Lemma 4(v): $\Theta^{-1}(\mathbb{R}) \cap \Omega_\delta = \mathbb{R}$, so $t_0 \in \mathbb{R}$. $\square$

6 The Riemann Hypothesis

Theorem 24 (Riemann Hypothesis)**.** *Every nontrivial zero ρ of ζ satisfies $\Re\rho = \frac{1}{2}$.*

Proof. Let $\rho = \beta + i\gamma$ be a nontrivial zero with $\beta \in (0,1)$, $\gamma \in \mathbb{R}$.

Reduction to $\beta \leq \frac{1}{2}$. The functional equation $\xi(s) = \xi(1-s)$ and reality of ζ on $\mathbb{R} \cap \{\Re s = \frac{1}{2}\}$ give the symmetry $\rho \mapsto 1 - \bar{\rho}$: if $\rho = \beta + i\gamma$ is a zero then so is $1 - \bar{\rho} = (1-\beta) + i\gamma$. So it suffices to establish $\beta = \frac{1}{2}$ under the assumption $\beta \in (0, \frac{1}{2}]$; the case $\beta \in (\frac{1}{2}, 1)$ follows by applying the result to $1 - \bar{\rho}$, whose real part $1 - \beta$ lies in $(0, \frac{1}{2})$.

Assume henceforth $\beta \in (0, \frac{1}{2}]$ and define

$$\delta_\rho := \tfrac{1}{2}\min(\beta, 1-\beta) = \tfrac{\beta}{2} \in (0, \tfrac{1}{4}], \qquad t_\rho := \gamma + i(\tfrac{1}{2} - \beta).$$

Then $\Im t_\rho = \frac{1}{2} - \beta \geq 0$ and $|\Im t_\rho| = \frac{1}{2} - \beta = \frac{1}{2} - 2\delta_\rho < \frac{1}{2} - \delta_\rho$, so $t_\rho \in \Omega_{\delta_\rho}$.

Choose any δ_ρ-admissible $c > c_{**}(\delta_\rho)$ (Definition 3). By the strip factorization (Theorem 21) on Ω_{δ_ρ},

$$Z(t_\rho) = e^{i\vartheta(t_\rho)}\zeta(\tfrac{1}{2} + it_\rho) = e^{i\vartheta(t_\rho)}\zeta(\rho) = 0.$$

By Corollary 23, $t_\rho \in \mathbb{R}$, so $\Im t_\rho = \frac{1}{2} - \beta = 0$, hence $\beta = \frac{1}{2}$. $\square$

References

[1] N. I. Akhiezer, *Theory of Approximation*, Ungar, 1956.

[2] M. V. Berry, The Riemann–Siegel expansion for the zeta function: high orders and remainders, *Proc. Roy. Soc. A* **450** (1995), 439–462.

[3] S. Bochner and R. S. Phillips, Absolutely convergent Fourier expansions for non-commutative normed rings, *Ann. Math.* (2) **43** (1942), 409–418.

[4] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.

[5] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann–Siegel-Formel*, Dissertation, Georg-August-Universität Göttingen, 1979.

[6] B. Ya. Levin, *Lectures on Entire Functions*, Transl. Math. Monogr. **150**, AMS, 1996.

[7] F. W. J. Olver, *Asymptotics and Special Functions*, Academic Press, 1974.

[8] H. Omer and B. Torresani, Time-frequency and time-scale analysis of deformed stationary processes, HAL preprint hal-01094835, 2017.

[9] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., OUP, 1986.

[10] N. Wiener, Generalized harmonic analysis, *Acta Math.* **55** (1930), 117–258.

[11] A. M. Yaglom, *Correlation Theory of Stationary and Related Random Functions*, Vol. I, Springer, 1987.